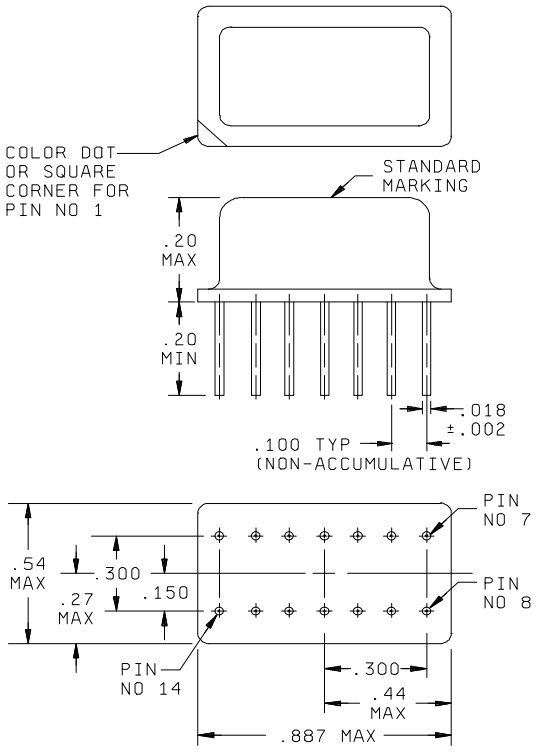
INCH-POUND
MIL-PRF-55310/26A
25 March 1998
SUPERSEDING
MIL-O-55310/26
19 July 1990

PERFORMANCE SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 1 (CRYSTAL OSCILLATOR (XO)), 10 kHz THROUGH 65 MHz, HERMETIC SEAL, SQUARE WAVE, HIGH SPEED CMOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55310.



Pin number	Function
1	NC
2	NC
3	NC
4	NC
5	NC
6	NC
7	GND/case
8	OUTPUT
9	NC
10	NC
11	NC
12	NC
13	NC
14	B+

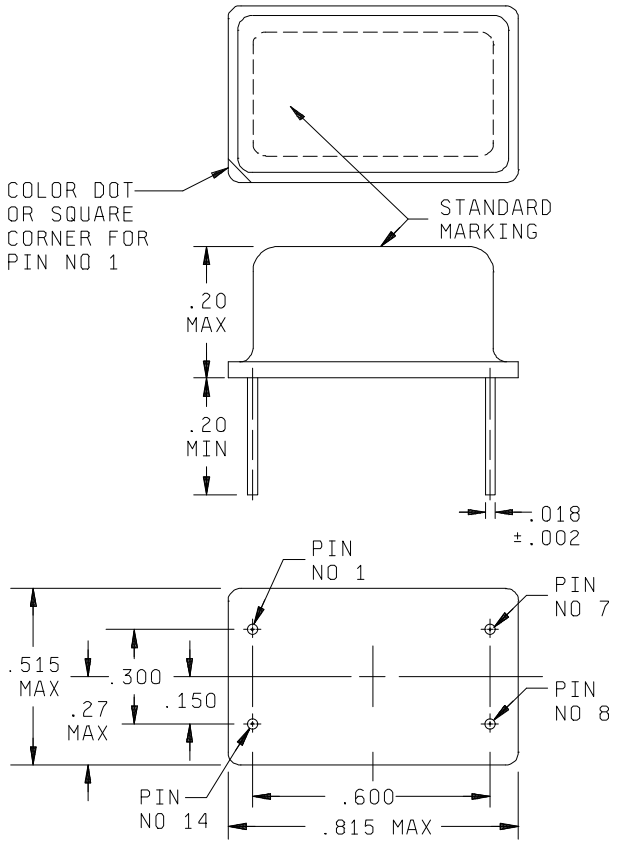
Inches	mm
.002	0.05
.018	0.46
.100	2.54
.150	3.81
.20	5.1
.27	6.9
.300	7.62
.44	11.2
.54	13.7
.887	22.53

Configuration A

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Pin number	Function
1	NC
7	GND/case
8	Output
14	B+

Inches	mm
.002	0.05
.018	0.46
.150	3.81
.20	5.1
.27	6.9
.300	7.62
.515	13.08
.600	15.24
.815	20.70

Configuration B

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. All pins with the NC function may be connected internally and are not to be used as external tie points or connections.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Package configuration: See figure 1.

Terminals: See figure 1.

Weight: 0.5 ounces (14.2 grams-mass), maximum.

Package: Class 2.

Calibration: Manufacturer calibrated.

Screening: In accordance with MIL-PRF-55310, class B or S, as applicable.

Temperature:

Operating: See table I.

Storage: -62°C to +125°C.

Load test circuit: See figure 2.

Seal: Hermetic in accordance with MIL-PRF-55310, maximum leakage rate 5 x 10^{-8} atm cc/s.

Supply voltage (B+): -5.0 V dc ±10 percent.

Overvoltage survivability: In accordance with MIL-PRF-55310.

Supply current: At designated supply voltage (see table I).

Start-up time: 15 milliseconds maximum, measurement shall be taken at reference temperature and operating temperature range endpoints.

Nominal output frequency: Frequency as designated at time of acquisition (see table I).

Output logic voltage levels at designated HCMOS load: See figure 3.

Logic 1: 90 percent of V_{DD}, minimum.

Logic 0: 10 percent of V_{DD}, maximum.

Output waveform: Symmetrical square wave, HCMOS logic compatible (see figure 3).

Duty cycle: See table I and figure 3.

Rise and fall times (see table I): Measurements shall be taken at the 10 percent and 90 percent peak-to-peak output voltage levels, with peak-to-peak output defined as Logic 1 - Logic 0 (see figure 3).

Initial accuracy at reference temperature: See table I.

Initial frequency-temperature accuracy (1/2 temperature cycle): See table I. Measurements shall be taken at ten equally spaced increments over the specified operating temperature range.

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TABLE I. Dash numbers and operating characteristics.

Dash number		Output frequency range	Input current	Pulse characteristics 1/		Initial accuracy	Frequency aging per	Initial frequency 2/ temperature accuracy		
Config-uration			(max) at 5.0 V ±10% 3/	Rise/fall time (max)	Duty cycle min-max	at 23°C ±1°C 2/	year (max) 4/	-55°C to +125°C	-55°C to +105°C	-20°C to +70°C
A	B							A	B	C
					percent	ppm	ppm	ppm	ppm	ppm
02	03	.01 MHz to 1.0 MHz	10 mA	10 ns	45 to 55	±15	±5	±65	±55	±40
06	07	.01 MHz to 1.0 MHz	10 mA	10 ns	45 to 55	±25	±10	±100	±75	±50
22	23	1 MHz to 4 MHz	15 mA	10 ns	45 to 55	±15	±5	±65	±55	±40
26	27	1 MHz to 4 MHz	15 mA	10 ns	45 to 55	±25	±10	±100	±75	±50
32	33	4 MHz to 20 MHz	20 mA	10 ns	40 to 60	±15	±5	±65	±55	±40
36	37	4 MHz to 20 MHz	20 mA	10 ns	40 to 60	±25	±10	±100	±75	±50
42	43	20 MHz to 35 MHz	35 mA	10 ns	40 to 60	±15	±5	±65	±55	±40
46	47	20 MHz to 35 MHz	35 mA	10 ns	40 to 60	±25	±10	±100	±75	±50
52	53	35 MHz to 50 MHz	40 mA	5 ns	40 to 60	±15	±5	±65	±55	±40
56	57	35 MHz to 50 MHz	40 mA	5 ns	40 to 60	±25	±10	±100	±75	±50
62	63	50 MHz to 65 MHz	70 mA	5 ns	40 to 60	±15	±5	±65	±55	±40
66	67	50 MHz to 65 MHz	70 mA	5 ns	40 to 60	±25	±10	±100	±75	±50

1/ See figure 3.
2/ Up to 30 days following shipment, see table II.
3/ No load condition.
4/ After 30 days following shipment, see table II.

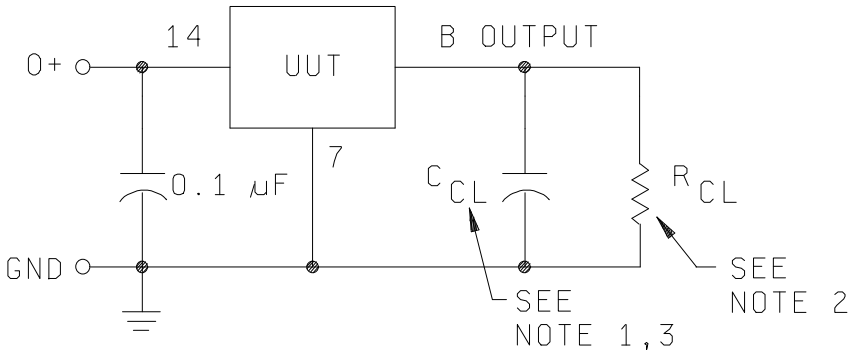
TABLE II. Frequency aging limits.

	5 ppm per year 1/	10 ppm per year 1/
Maximum change over 30 days	±0.7 ppm	±1.5 ppm
Projected maximum change for 1 year after 30 days	±5.0 ppm	±10.0 ppm

1/ See table I.

Frequency-voltage tolerance ±2 ppm maximum for a ±10 percent change in supply voltage. Measurements shall be taken at reference temperature and operating temperature range endpoints.

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NOTES:
1. For HCMOS: C_{CL} = 15 pF ±5 percent.
2. For HCMOS: R_{CL} = 10 kΩ ±5 percent.
3. C_{CL} includes scope capacitance.

FIGURE 2. Load test circuit.

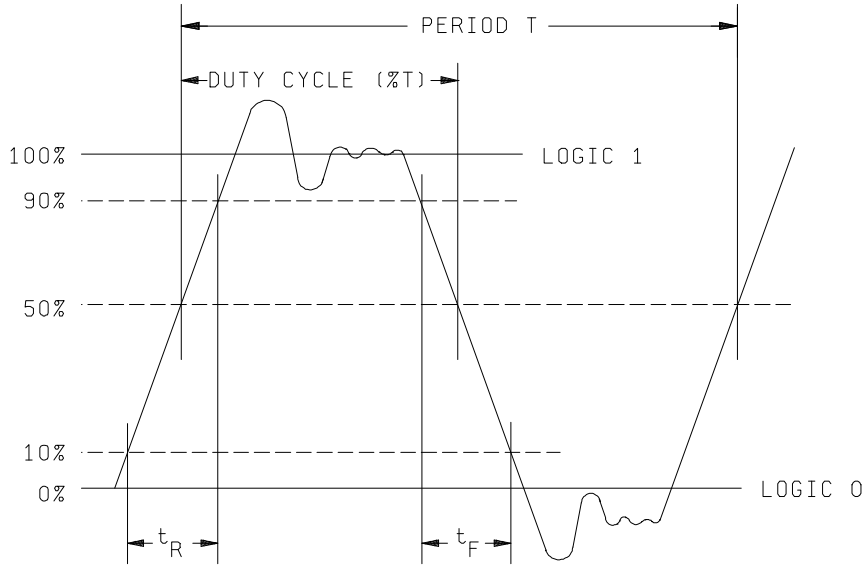


FIGURE 3. Waveform.

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Frequency-environmental tolerance: ±3 ppm.

Frequency aging: Frequency aging shall be in accordance with MIL-PRF-55310 and shall meet the limits of table II.

Vibration, sinusoidal: Method 204 of MIL-STD-202.

Nonoperating: Test condition D.

Operating: Not required.

Shock (nonoperating): Method 213 of MIL-STD-202, test condition I.

Thermal shock (nonoperating): Method 107 of MIL-STD-202, test condition B.

Ambient pressure:

Nonoperating: In accordance with MIL-PRF-55310.

Operating: Method 105 of MIL-STD-202, test condition C.

Resistance to soldering heat: Method 210 of MIL-STD-202, test condition E.

Moisture resistance: Method 106 of MIL-STD-202.

Terminal strength: Method 211 of MIL-STD-202, test condition C.

Applied force: 2 pounds each terminal for 10 seconds.

Bends: 5 at 45 degrees each.

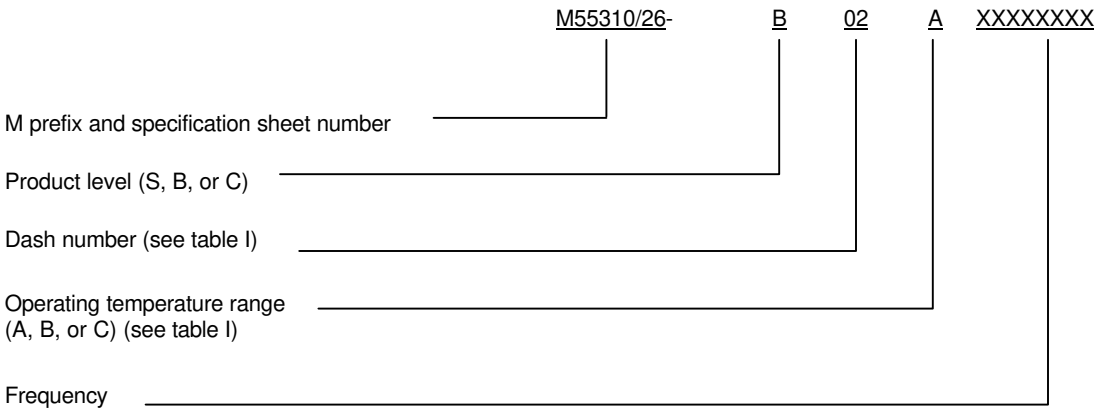
Solderability: Method 208 of MIL-STD-202.

Resistance to solvents: Method 215 of MIL-STD-202.

Screening: In accordance with MIL-PRF-55310, class B or S, as applicable.

Part or Identifying Number (PIN): Consists of “M” prefix followed by specification sheet number, a dash and coded alphas, and numeric number. See example:

EXAMPLE



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Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 11, 17, 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0713)

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